Chapter 14

Nonsmooth Optimization

14.1 Generalized Gradients

In this book, nonsmooth functions are those functions which need not be differentiable. Therefore they are also called nondifferentiable functions.

The nonlinear programming problem (8.1.1)–(8.1.3) is said to be a nonsmooth optimization problem, provided that either the objective function $f(x)$ or at least one of the constraint functions $c_i(x), (i = 1, \cdots, m)$ is a nonsmooth function.

To conclude the book, we would like to give an initial and readable introduction to nonsmooth optimization. To study the optimality condition of nonsmooth optimization and construct some numerical methods for solving nonsmooth optimization problems, we first introduce the fundamental conceptions and properties of nonsmooth functions.

Let X be a Banach space with a norm $\| \cdot \|$ defined on X. Let Y be a subset of X. A function $f : Y \to R$ is Lipschitz on Y if $f(x)$ satisfies

$$|f(x) - f(y)| \leq K\|x - y\|, \ \forall x, y \in Y \subseteq X, \tag{14.1.1}$$

where K is called the Lipschitz constant. The inequality (14.1.1) is also referred to as a Lipschitz condition.

We define a generalized sphere

$$B(x, \epsilon) = \{y \mid \|x - y\| \leq \epsilon\}. \tag{14.1.2}$$

We say that f is Lipschitz near x if, for some $\epsilon > 0$, f satisfies a Lipschitz condition on $B(x, \epsilon)$.

It is easy to see that a function having a Lipschitz property near a point need not be differentiable there, nor need admit a directional derivative in the classical sense.

The directional derivative of f at x in the direction d is

$$f'(x;d) = \lim_{t\downarrow 0} \frac{f(x+td)-f(x)}{t}. \tag{14.1.3}$$

The upper Dini directional derivative of f at x in the direction d is

$$f^{(D)}(x;d) = \limsup_{t\downarrow 0} \frac{f(x+td)-f(x)}{t}. \tag{14.1.4}$$

Let f be Lipschitz near a given point x, and let d be any other vector in X. The generalized directional derivative of f at x in the direction d is defined as follows:

$$f^o(x;d) = \limsup_{\substack{y\to x\\ t\downarrow 0}} \frac{f(y+td)-f(y)}{t}, \tag{14.1.5}$$

where of course y is a vector in X and t is a positive scalar, and $t \downarrow 0$ denotes that t tends to zero monotonically and downward. Since the generalized directional derivative is due to Clarke [60], it is also referred to as a Clarke directional derivative.

For a locally Lipschitz function, the directional derivative may not exist but the Dini and the Clarke directional derivatives always exist. Obviously, we always have the relation

$$f^{(D)}(x;d) \le f^o(x;d) \tag{14.1.6}$$

for all x and d. If the directional derivative exists, then it is equal to the upper Dini directional derivative. If $f'(x;d)$ exists at x for all d, then f is said to be directionally differentiable at x. If f is directionally differentiable at x and

$$f'(x;d) = f^o(x;d), \tag{14.1.7}$$

then f is said to be regular at x. The function f is said to be a regular function if it is regular everywhere.

Lemma 14.1.1 *Let $f(x)$ be Lipschitz near x. Then*

1. *The function $d \to f^o(x;d)$ is positive homogeneous and subadditive on X, and satisfies*

$$|f^o(x;d)| \le K\|d\|. \tag{14.1.8}$$

2. *$f^o(x;d)$ is Lipschitz on X as a function of d.*

3. *$f^o(x;d)$ is upper semicontinuous as a function of $(x;d)$.*

4. *$f^o(x;-d) = (-f)^o(x;d)$.*

Proof. 1) In view of (14.1.1), (14.1.5) and the fact that $f(x)$ is Lipschitz near x, we immediately have (14.1.8). The fact that

$$f^o(x;\lambda d) = \lambda f^o(x;d)$$

for any $\lambda > 0$ is immediate from the definition (14.1.5). Now we turn to the subadditivity.

From (14.1.5), we have

$$\begin{aligned} f^o(x;d_1+d_2) &= \limsup_{\substack{y\to x \\ t\downarrow 0}} \frac{f(y+t(d_1+d_2))-f(y)}{t} \\ &\leq \limsup_{\substack{y\to x \\ t\downarrow 0}} \frac{f(y+td_1+td_2)-f(y+td_2)}{t} \\ &\quad + \limsup_{\substack{y\to x \\ t\downarrow 0}} \frac{f(y+td_2)-f(y)}{t} \\ &\leq f^o(x;d_1)+f^o(x;d_2). \end{aligned} \tag{14.1.9}$$

2) Let any $d_1, d_2 \in X$ be given. We have from the Lipschitz condition that

$$f(y+td_1)-f(y) \leq f(y+td_2)-f(y)+Kt\|d_1-d_2\| \tag{14.1.10}$$

holds for y near x, $t > 0$ sufficiently small. Dividing by t and taking upper limits as $y \to x, t \downarrow 0$, gives

$$f^o(x;d_1) \leq f^o(x;d_2)+K\|d_1-d_2\|. \tag{14.1.11}$$

Similarly, we obtain

$$f^o(x;d_2) \leq f^o(x;d_1)+K\|d_1-d_2\|. \tag{14.1.12}$$

The above two inequalities give

$$|f^o(x;d_1)-f^o(x;d_2)| \leq K\|d_1-d_2\|. \tag{14.1.13}$$

Then we complete 2).

3) Now let $\{x_i\}$ and $\{d_i\}$ be arbitrary sequences with $x_k \to x$ and $d_k \to d$ respectively. For each i, by definition of upper limit, there exist $y_k \in X$ and $t_k > 0$ such that

$$\|y_k - x_k\| + t_k < \frac{1}{k}, \tag{14.1.14}$$

$$\begin{aligned} & f^o(x_k; d_k) - \frac{1}{k} \\ \leq\ & \frac{f(y_k + td_k) - f(y_k)}{t_k} \\ \leq\ & \frac{f(y_k + t_k d_k) - f(y_k + t_k d)}{t_k} + \frac{f(y_k + t_k d) - f(y_k)}{t_k}. \end{aligned} \tag{14.1.15}$$

Upon taking upper limits (as $k \to \infty$), we derive

$$\limsup_{k\to\infty} f^o(x_k; d_k) \leq f^o(x; d), \tag{14.1.16}$$

which establishes the upper semicontinuity.

4) Finally, we calculate

$$\begin{aligned} f^o(x; -d) &= \limsup_{\substack{y\to x \\ t\downarrow 0}} \frac{f(y - td) - f(y)}{t} \\ &= \limsup_{\substack{u\to x \\ t\downarrow 0}} \frac{(-f)(u + td) - (-f)(u)}{t} \\ &= (-f)^o(x; d), \end{aligned} \tag{14.1.17}$$

where $u = y - td$. Hence, we complete the proof. $\Box$

The Hahn-Banach Theorem (for example, see Cryer [72], Theorem 7.4) asserts that any positive homogeneous and subadditive functional on X majorizes some linear functional on X. Under the condition of Lemma 14.1.1, therefore, there is at least one linear functional $\xi : X \to R$ such that, for all $d \in X$, one has

$$f^o(x; d) \geq \xi(d).$$

It follows also that ξ is bounded, and hence belongs to the dual space X^* of continuous linear functionals on X, for which we adopt the convention of using $\langle \xi, d\rangle$ or $\langle d, \xi\rangle$ for $\xi(d)$.

We then give the following definition:

Definition 14.1.2 *Let $f(x)$ be Lipschitz near x. Then we say that the generalized differential (or Clarke differential) of f at x is the set*

$$\partial f(x) = \{\xi \in X^* \mid f^o(x; d) \geq \langle \xi, d\rangle, \forall d \in X\}. \tag{14.1.18}$$

The ξ is said to be the generalized gradient.

The norm $\|\xi\|_*$ in conjugate space X^* is defined as

$$\|\xi\|_* = \sup\{\langle \xi, d\rangle \ : \ d \in X, \|d\| \leq 1\}. \tag{14.1.19}$$

The following summarizes some basic properties of the generalized gradient.

Lemma 14.1.3 *Let $f(x)$ be Lipschitz near x. Then*

1) $\partial f(x)$ is a nonempty, convex, weak-compact subset of X^* and $\|\xi\|_* \leq K$ for every $\xi \in \partial f(x)$.*

2) For every $d \in X$, one has

$$f^o(x; d) = \max_{\xi \in \partial f(x)} \{\langle \xi, d\rangle\}. \tag{14.1.20}$$

Proof. Assertion 1) is immediate from the preceding remarks and Lemma 14.1.1. (The weak*-compactness follows from Alaoglu's Theorem.)

Assertion 2) is simply a restatement of the fact that $\partial f(x)$ is by definition the weak*-closed convex set whose support function is $f^o(x; \cdot)$. To see this independently, suppose that for some d, $f^o(x; d)$ exceeded the given maximum (it cannot be less, by definition of $\partial f(x)$). According to a common version of the Hahn-Banach Theorem there is a linear functional ξ majorized by $f^o(x, \cdot)$ and agreeing with it at d. It follows that $\xi \in \partial f(x)$, whence $f^o(x; d) > \langle \xi; d\rangle = f^o(x; d)$. This contradiction establishes the assertion 2). □

Note that if $f(x)$ is convex, the conceptions of generalized directional derivative and generalized gradient coincide with that of directional derivative and subgradient defined for convex functions due to Rockafellar [288].

As an example, we calculate the generalized differential of the absolute-value function in the case of $X = R$.

Consider the problem

$$f(x) = |x|.$$

Obviously, f is Lipschitz by the triangle inequality. If $x > 0$, we calculate

$$f^o(x;d) = \limsup_{\substack{y\to x\\ t\downarrow 0}} \frac{y+td-y}{t} = d,$$

so that

$$\partial f(x) = \{\xi \mid d \geq \xi d,\ \forall d \in R\}$$

reduces to the singleton $\{1\}$.

Similarly, we have

$$\partial f(x) = \{-1\} \ \text{ if } \ x < 0.$$

The remaining case is $x = 0$. We find

$$f^o(0;d) = \begin{cases} d, & \text{if } d \geq 0, \\ -d, & \text{if } d < 0, \end{cases}$$

that is

$$f^o(0,d) = |d|.$$

Thus $\partial f(0)$ consists of those ξ satisfying $|d| \geq \xi d$ for all d; that is $\partial f(0) = \{-1, 1\}$. Therefore, we conclude

$$\partial f(x) = \begin{cases} \{1\}, & x > 0, \\ \{-1\}, & x < 0, \\ \{-1,1\}, & x = 0. \end{cases}$$

We introduce an important conception as follows.

The support function of a nonempty subset Ω of X is a function $\sigma_\Omega(\xi) : X^* \to R \cup \{+\infty\}$ defined by

$$\sigma_\Omega(\xi) := \sup_{x\in\Omega}\{\langle \xi, x\rangle\}. \tag{14.1.21}$$

It is easy to see that $f^o(x;\cdot)$ is the support function of $\partial f(x)$.

By (14.1.21) and Definition 14.1.2, the following lemma is obvious.

Lemma 14.1.4 *Let $f(x)$ be Lipschitz near x. Then*

$$\xi \in \partial f(x) \textit{ if and only if } f^o(x;d) \geq \langle \xi; d\rangle\ \forall d \in X. \tag{14.1.22}$$

Furthermore, $\partial f(x)$ has the following properties:

a)

$$\partial f(x) = \cap_{\delta>0} \cup_{y\in x+B(0,\delta)} \partial f(y), \tag{14.1.23}$$

where

$$B(0,\delta) = \{x \mid \|x\| \le \delta,\, x \in X\}.$$

If X is finite-dimensional, then the ∂f is upper semi-continuous.

b) If $f_i\,(i = 1, \cdots, m)$ are finitely many Lipschitz functions near x, then $\sum_{i=1}^m f_i$ is also Lipschitz near x and

$$\partial(\sum_{i=1}^m f_i)(x) \subset \sum_{i=1}^m \partial f_i(x). \tag{14.1.24}$$

c) If $f(x) = g(h(x))$, where $h(x) = (h_1(x), \cdots, h_n(x))^T$, each $h_i(x)$ is Lipschitz near x, and $g(x)$ is Lipschitz near $h(x)$, then $f(x)$ is Lipschitz near x and

$$\partial f(x) \subset \overline{co}\left\{\sum_{i=1}^n \alpha_i\xi_i :\ \xi_i \in \partial h_i(x), \alpha \in \partial g(h)|_{h=h(x)}\right\}, \tag{14.1.25}$$

where $\overline{co}$ denotes a weak*-compact convex hull (see Theorem 2.3.9 in Clarke [60]).

Below, we turn to the optimal condition for minimization of a Lipschitz function. By Lemma 14.1.4, we can immediately deduce the first-order necessary condition.

Theorem 14.1.5 *If $f(x)$ attains a local minimum or maximum at x^* and $f(x)$ is Lipschitz near x^*, then*

$$0 \in \partial f(x^*). \tag{14.1.26}$$

Proof. If x^* is a local minimizer of $f(x)$, then it follows from the definition (14.1.5) that for any $d \in X$ we have

$$f^o(x^*; d) \ge 0. \tag{14.1.27}$$

Thus, by Lemma 14.1.4, we have $0 \in \partial f(x^*)$.

If x^* is a local maximizer of $f(x)$, then x^* is a local minimizer of $(-f)(x)$. It suggests that $0 \in \partial(-f)(x^*)$. It is not difficult to show that for any scalar

s, one has $\partial(sf)(x) = s\partial f(x)$. Therefore $0 \in \partial(-f)(x^*) = -\partial f(x^*)$ which means $0 \in \partial f(x^*)$. Hence we complete the proof. $\square$

A point x^* is called a stationary point of f if f is directionally differentiable at x^* and for all d,

$$f'(x^*, d) \geq 0. \tag{14.1.28}$$

A point x^* is called a Dini stationary point of f if for all d,

$$f^{(D)}(x^*; d) \geq 0. \tag{14.1.29}$$

A point x^* is called a Clarke stationary point of f if for all d,

$$f^o(x^*; d) \geq 0, \tag{14.1.30}$$

i.e.,

$$0 \in \partial f(x^*). \tag{14.1.31}$$

A local minimizer x^* of a local Lipschitzian function f is always a Dini stationary point of f. If f is directionally differentiable at x^*, then x^* is also a stationary point. A Dini stationary point is always a Clarke stationary point but not vice versa.

Now we state the sufficient condition which is based on a lemma below.

Lemma 14.1.6 *Let $f(x)$ be convex and Lipschitz near x^*, then the generalized differential $\partial f(x)$ coincides with the subdifferential at x, and the generalized directional derivative $f^o(x; d)$ coincides with the directional derivative $f'(x; d)$ for each d.*

Proof. It is known from convex analysis that $f'(x; d)$ exists for each d and $f'(x; d)$ is the support function of the subdifferential at x. It suffices therefore to prove that for any d, $f^o(x; d) = f'(x; d)$. Note that

$$f^o(x; d) = \lim_{\epsilon \downarrow 0} \sup_{\|x' - x\| < \epsilon\delta} \sup_{0 < t < \epsilon} \frac{f(x' + td) - f(x')}{t}, \tag{14.1.32}$$

where δ is any fixed positive number. It follows from the definition of convex function that the function

$$t \to \frac{f(x' + td) - f(x')}{t}$$

is non-decreasing, whence

$$f^o(x;d) = \lim_{\epsilon \downarrow 0} \sup_{\|x'-x\|<\epsilon\delta} \frac{f(x'+\epsilon d) - f(x')}{\epsilon}.$$

Now by the Lipschitz condition, for any x' in $x + B(0, \epsilon\delta)$, one has

$$\left| \frac{f(x'+\epsilon d) - f(x')}{\epsilon} - \frac{f(x+\epsilon d) - f(x)}{\epsilon} \right| \leq 2\delta K,$$

so that

$$f^o(x;d) \leq \lim_{\epsilon \downarrow 0} \frac{f(x+\epsilon d) - f(x)}{\epsilon} + 2\delta K = f'(x;d) + 2\delta K.$$

Since δ is arbitrary, we deduce $f^o(x;d) \leq f'(x;d)$. Therefore the equality follows. The proof is complete. $\square$

We now can state the sufficient condition.

Theorem 14.1.7 *Let $f(x)$ be convex and Lipschitz near x^*, and*

$$0 \in \partial f(x^*), \tag{14.1.33}$$

then x^* is a local minimizer of $f(x)$.

Proof. For a convex and Lipschitzian function, from Lemma 14.1.6, the generalized differential and the subdifferential

$$\{\xi \in X^* \mid f(z) - f(x) \geq \langle \xi, z - x \rangle, \ \forall z \in X\} \tag{14.1.34}$$

are equivalent. Then, by (14.1.33) and (14.1.34), we have that x^* is a local minimizer of $f(x)$. $\square$

Hence, for a convex and Lipschitzian function, (14.1.33) is a sufficient and necessary condition for x^* to be a local minimizer of $f(x)$. This is also equivalent to

$$f^o(x^*;d) \geq 0, \ \forall d \in X. \tag{14.1.35}$$

For a convex and Lipschitzian function, the generalized directional derivative $f^o(x;d)$ coincides with the directional derivative $f'(x;d)$:

$$f'(x;d) = \lim_{t \downarrow 0} \frac{f(x+td) - f(x)}{t}. \tag{14.1.36}$$

(We would like to mention that, from convex analysis, convex functions are Lipschitz except in the pathological case).

Furthermore, we can state a sufficient condition for a strict (strong) minimizer.

Theorem 14.1.8 *Let $f(x)$ be convex and Lipschitz near x^*. If*

$$f'(x^*; d) > 0, \ \forall d \neq 0, d \in X, \tag{14.1.37}$$

then x^* is a strict (strong) minimizer of $f(x)$, i.e., there exists $\delta > 0$ such that

$$f(x) - f(x^*) \geq \delta \|x - x^*\| \tag{14.1.38}$$

holds for all x sufficiently close to x^*.

Proof. Define a set

$$S = \{d \,|\, d \in X, \|d\| = 1\}.$$

Obviously, S is compact and closed. By (14.1.37), it follows that $f'(x^*, d)$ is positive on S. Then, from the continuity of $f'(x^*, d)$ (in fact, $f'(x^*; d)$ is a positive homogeneous and convex function of d), there exists $\delta > 0$ such that

$$f'(x^*; d) \geq 2\delta, \ \forall d \in S. \tag{14.1.39}$$

Then for any $d \in S$, there exists $t(d) > 0$ such that

$$f(x^* + td) - f(x^*) \geq td, \ \forall t \in [0, t(d)]. \tag{14.1.40}$$

By convexity and continuity of $f(x)$, we can show that there is an $\epsilon > 0$ such that

$$t(d) \geq \epsilon, \ \forall d \in S. \tag{14.1.41}$$

Hence, for all x with $\|x - x^*\| \leq \epsilon$, we have

$$f(x) - f(x^*) \geq \delta \|x - x^*\| \tag{14.1.42}$$

which indicates (14.1.38). $\Box$

However, for a non-convex function, the above sufficiency result is not true. In fact, let us consider an example below: for $f : R^1 \to R^1$,

$$f(x) = \begin{cases} (-1)^{k+1} \left[\frac{1}{2^{k+1}} - 3x\right], & x \in \left[\frac{1}{2^{k-1}}, \frac{1}{2^k}\right], \\ 0, & x = 0, \end{cases} \tag{14.1.43}$$

$$f(x) = f(-x), \ \forall x \in [-1, 0). \tag{14.1.44}$$

Clearly, $f(x)$ is Lipschitz on $[-1, 1]$, and

$$f^o(x^*; \pm 1) = 3 > 0 \tag{14.1.45}$$

at $x^* = 0$, which means there are two generalized directional derivatives equal to 3. But $x^* = 0$ is not the extreme point.

14.2 Nonsmooth Optimization Problem

Consider unconstrained optimization problem

$$\min_{x \in X} f(x), \tag{14.2.1}$$

where $f(x)$ is a nondifferentiable function defined in Banach space and satisfies a Lipschitz condition. From the discussion in §14.1, if x^* is a solution of (14.2.1), then

$$0 \in \partial f(x^*), \tag{14.2.2}$$

i.e., x^* is a stationary point of (14.2.1).

As to solution for nonsmooth optimization problem (14.2.1), there are two main difficulties if one is using a method suitable for differentiable problems. First, it is not easy to give a termination criteria. It is well-known that when x approaches the minimizer of a continuously differentiable function $f(x)$, the $\|\nabla f(x)\|$ is very small. Hence the common termination criteria

$$\|\nabla f(x)\| \leq \epsilon \tag{14.2.3}$$

is used. However, for a nonsmooth function, there are no similar results. For example, consider the simple problem that $f : R^1 \to R^1$ and $f(x) = |x|$. Then, for any x that is not a solution, $f(x)$ is differentiable and

$$|\partial f(x)| = |\nabla f(x)| = 1. \tag{14.2.4}$$

Hence, in this case, we cannot use (14.2.3) as a termination criteria.

Second, as indicated by Wolfe [354], when $f(x)$ is nondifferentiable, if one uses the steepest descent method with line search to solve (14.2.1), it is possible to generate a sequence $\{x_k\}$ converging to a non-stationary point. For example, let $f : R^2 \to R^1, x = (u, v)^T$ and

$$f(x) = \max\left[\frac{1}{2}u^2 + (v-1)^2, \frac{1}{2}u^2 + (v+1)^2\right]. \tag{14.2.5}$$

Suppose that x_k has the form

$$x_k = \begin{pmatrix} 2(1+|\epsilon_k|) \\ \epsilon_k \end{pmatrix}, \tag{14.2.6}$$

where $\epsilon_k \neq 0$. Then we can calculate

$$\nabla f(x_k) = \begin{pmatrix} 2(1+|\epsilon_k|) \\ 2(1+|\epsilon_k|)t_k \end{pmatrix} = 2(1+|\epsilon_k|)\begin{pmatrix} 1 \\ t_k \end{pmatrix}, \tag{14.2.7}$$

where $t_k = \text{sign}(\epsilon_k)$. If we employ the negative gradient direction $-\nabla f(x_k)$, then we have

$$\begin{aligned} x_{k+1} &= x_k + \alpha_k(-\nabla f(x_k)) = \begin{bmatrix} 2(1+|\epsilon_k|/3) \\ -\epsilon_k/3 \end{bmatrix} \\ &= \begin{bmatrix} 2(1+|\epsilon_{k+1}|) \\ \epsilon_{k+1} \end{bmatrix}, \end{aligned} \tag{14.2.8}$$

where $\epsilon_{k+1} = -\epsilon_k/3 \neq 0$. Then we can prove $\epsilon_k \to 0$. So, for a given initial point as $(2+2|\delta|, \delta)^T$, where $\delta \neq 0$, the sequence generated by the steepest descent method with exact line search converges to $(2,0)^T$. It is obvious that $(2,0)^T$ is not the stationary point.

A nonsmooth constrained optimization problem has the form

$$\min_{x\in Y} f(x), \tag{14.2.9}$$

where $Y \subseteq X$ is a set, or a feasible region. Define a distance function

$$\text{dist}(x, Y) = \min_{y\in Y} \|y - x\|. \tag{14.2.10}$$

By the theory of penalty function, under suitable conditions, (14.2.9) is equivalent to

$$\min_{x\in X} f(x) + \sigma\text{dist}(x, Y), \tag{14.2.11}$$

where $f(x) + \sigma\text{dist}(x, Y)$ is a non-differentiable function. Hence, the nonsmooth constrained optimization problem is transformed to an equivalent nonsmooth unconstrained problem. This interprets why one always is interested in studying nonsmooth unconstrained optimization problems.

There are many examples of nonsmooth optimization problems, for example, the minimax problem

$$\min_{x\in X} \max_{1\leq i\leq m} f_i(x). \tag{14.2.12}$$

In addition, in order to solve nonlinear equations

$$f_i(x) = 0, i = 1, \cdots, m, \tag{14.2.13}$$

we often find the solution of the minimization problem

$$\min_{x\in X} f(x) = \min_{x\in X} \|\bar{f}(x)\| \tag{14.2.14}$$

under some norm $\|\cdot\|$, where $f(x) = \|\bar{f}(x)\|$, $\bar{f}(x) = (f_1(x), \cdots, f_m(x))$ is a vector function from X to R^n. Clearly, the problem (14.2.14) is a nonsmooth optimization problem. In particular, if $\|\cdot\| = \|\cdot\|_1$, it is a L_1 minimization problem; if $\|\cdot\| = \|\cdot\|_\infty$, it is Chebyshev approximation problem.

Note that the exact penalty function (10.6.2) is also a nonsmooth function. Therefore, the minimization to the exact penalty function is also a nonsmooth optimization problem.

14.3 The Subgradient Method

The subgradient method is a direct generalization of the steepest descent method, which generates a sequence $\{x_k\}$ by use of $-g_k$ as a direction, where $g_k \in \partial f(x_k)$.

Let $f(x)$ be a convex function on R^n and the minimization problem be $\min_{x\in R^n} f(x)$. We have seen that the convex function is differentiable almost everywhere, and

$$\partial f(x) = \operatorname{conv} \Omega(x), \tag{14.3.1}$$

where convΩ denotes the convex hull of Ω,

$$\Omega(x) = \{g \mid g = \lim \nabla f(x_i), x_i \to x, \nabla f(x_i) \text{ exists}\}. \tag{14.3.2}$$

The subgradient method is described as follows.

Algorithm 14.3.1 *(The subgratient method)*

Step 1. Given an initial point $x_1 \in R^n$, $k := 1$.

Step 2. Compute $f(x_k), g_k \in \partial f(x_k)$.

Step 3. Choose stepsize $\alpha_k > 0$ and set

$$x_{k+1} = x_k - \alpha_k g_k / \|g_k\|_2, \tag{14.3.3}$$

$k := k+1$, go to Step 2. $\Box$

As shown in the above section, in the subgradient method, the exact line search may cause convergence to a non-stationary point.

In smooth optimization, inexact line search is to find the stepsize α_k such that

$$f(x_k + \alpha_k d_k) \le f(x_k) + \alpha_k c_1 d_k^T \nabla f(x_k), \tag{14.3.4}$$

where $c_1 \in (0,1)$ is a constant. For the steepest descent method, the above rule becomes

$$f(x_k - \alpha_k \nabla f(x_k)) \le f(x_k) - \alpha_k c_1 \|\nabla f(x_k)\|^2. \tag{14.3.5}$$

However, when $f(x)$ is nonsmooth, then for any $c_1 \in (0,1)$ and $g_k \in \partial f(x_k)$, the inequality

$$f(x_k - \alpha g_k) \le f(x_k) - \alpha c_1 \|g_k\|^2 \tag{14.3.6}$$

may not hold for any $\alpha > 0$. Therefore, inexact line search is also not practicable in nonsmooth optimization.

Note that a constant stepsize is unsuitable because the function may be nondifferentiable at the solution and then $\{g_k\}$ does not necessarily tend to zero, even if $\{x_k\}$ converges to the optimal point.

Therefore, the rules for determining α_k for the subgradient method are entirely different from that for the steepest descent method.

Although the exact and inexact line search for smooth optimization cannot be simply generalized to the nonsmooth case, the negative subgradient direction is a "good" direction such that the new iterate is closer to the solution.

Lemma 14.3.2 *Let $f(x)$ be a convex function and the set*

$$S^* = \{x \mid f(x) = f^* = \min_{x \in R^n} f(x)\} \tag{14.3.7}$$

be nonempty. If $x_k \notin S^*$, then for any $x^* \in S^*$ and $g_k \in \partial f(x_k)$, there must exist $T_k > 0$ such that

$$\left\| x_k - \alpha \frac{g_k}{\|g_k\|_2} - x^* \right\|_2 < \|x_k - x^*\|_2 \tag{14.3.8}$$

holds for all $\alpha \in (0, T_k)$.

Proof. For any x_k,

$$\begin{aligned} \left\| x_k - \alpha \frac{g_k}{\|g_k\|_2} - x^* \right\|_2^2 &= \|x_k - x^*\|_2^2 \\ &\quad + 2\alpha \left(\frac{g_k}{\|g_k\|_2} \right)^T (x^* - x_k) + \alpha^2. \end{aligned} \tag{14.3.9}$$

Since $g_k \in \partial f(x_k)$ and $x_k \notin S^*$, then we have

$$g_k^T(x^* - x_k) \le f(x^*) - f(x_k) < 0. \tag{14.3.10}$$

Define

$$T_k = -2g_k^T(x^* - x_k)/\|g_k\|_2 > 0, \tag{14.3.11}$$

then (14.3.9) becomes

$$\|x_k - \alpha\frac{g_k}{\|g_k\|_2} - x^*\|_2^2 = \|x_k - x^*\|_2^2 + \alpha(\alpha - T_k). \tag{14.3.12}$$

If $0 < \alpha < T_k$, then $\alpha(\alpha - T_k) < 0$ and further (14.3.8) holds. $\square$

By use of the above property of subgradient direction, we can take a sufficiently small step, such that the sequence $\{x_k\}$ is closer and closer to the solution. From the above lemma we can deduce easily the following result due to Shor [309].

Theorem 14.3.3 *Let $f(x)$ be convex and the set S^* be nonempty. For any $\delta > 0$ there exists $r > 0$ such that if the subgradient Algorithm 14.3.1 is used with $\alpha_k \equiv \alpha \in (0, r)$ then we have*

$$\liminf_{k\to\infty} f(x_k) \le f^* + \delta. \tag{14.3.13}$$

Note that the choice of constant stepsize $\alpha_k \equiv \alpha$ may cause the algorithm not to converge. Ermoliev [118] and Polyak [254] suggest choosing α_k which would satisfy

$$\alpha_k > 0, \ \lim_{k\to\infty} \alpha_k = 0, \tag{14.3.14}$$

$$\sum_{k=1}^{\infty} \alpha_k = \infty, \tag{14.3.15}$$

and establish the following convergence theorem.

Theorem 14.3.4 *Let $f(x)$ be convex, and the set S^* be nonempty and bounded. If α_k satisfies (14.3.14) and (14.3.15), then the sequence $\{x_k\}$ generated by Algorithm 14.3.1 satisfies*

$$\lim_{k\to\infty} dist(x_k, S^*) = 0, \tag{14.3.16}$$

where $dist(x, S)$ is defined by (14.2.10).

Proof. Since $f(x)$ is convex, there exists continuous function $\delta(\epsilon)$ such that

$$f(x) \leq f^* + \epsilon \tag{14.3.17}$$

holds for any

$$\mathrm{dist}(x, S^*) \leq \delta(\epsilon), \tag{14.3.18}$$

where $\delta(\epsilon) > 0\,(\forall \epsilon > 0)$. For each k, we define

$$\epsilon_k = f(x_k) - f^* \geq 0. \tag{14.3.19}$$

If $\epsilon_k > 0$, then

$$\begin{aligned}
\|x_{k+1} - x^*\|^2 &= \|x_k - x^*\|^2 + \alpha_k^2 - 2\alpha_k (x_k - x^*)^T g_k / \|g_k\|_2 \\
&= \|x_k - x^*\|^2 + \alpha_k^2 - 2\delta(\epsilon_k)\alpha_k \\
&\quad -2\alpha_k \left[x_k - x^* - \delta(\epsilon_k) \frac{g_k}{\|g_k\|_2} \right]^T g_k / \|g_k\|_2 \\
&\leq \|x_k - x^*\|^2 + \alpha_k^2 - 2\delta(\epsilon_k)\alpha_k.
\end{aligned} \tag{14.3.20}$$

Hence

$$[\mathrm{dist}(x_{k+1}, S^*)]^2 - [\mathrm{dist}(x_k, S^*)]^2 \leq -\alpha_k [2\delta(\epsilon_k) - \alpha_k]. \tag{14.3.21}$$

Define $\delta(0) = 0$, then the above expression holds for every k. Summing both sides of (14.3.21) gives

$$\liminf_{k \to \infty} \delta(\epsilon_k) = 0. \tag{14.3.22}$$

Thus,

$$\liminf_{k \to \infty} \mathrm{dist}(x_k, S^*) = 0. \tag{14.3.23}$$

Suppose to the contrary that the theorem is not true. Then there exist a positive constant $\delta' > 0$ and infinitely many k such that

$$\mathrm{dist}(x_{k+1}, S^*) > \mathrm{dist}(x_k, S^*) \tag{14.3.24}$$

and

$$\epsilon_k > \delta' \tag{14.3.25}$$

hold. From (14.3.24) and (14.3.21), we deduce that

$$2\delta(\epsilon_k) < \alpha_k \tag{14.3.26}$$

holds for sufficiently large k. Clearly, (14.3.26) contradicts (14.3.25). This contradiction shows the theorem. $\Box$

The above theorem indicates that Algorithm 14.3.1 converges if α_k satisfies (14.3.14)–(14.3.15). However, for such chosen α_k, the algorithm does not converge rapidly. In fact, we have

$$\|x_k - x^*\| + \|x_{k+1} - x^*\| \geq \|x_k - x_{k+1}\| = \alpha_k. \tag{14.3.27}$$

Then, by (14.3.27) and (14.3.15), we have immediately that

$$\sum_{k=1}^{\infty} \|x_k - x^*\| = +\infty, \tag{14.3.28}$$

which shows that the sequence cannot converge R-linearly.

To make the algorithm converge R-linearly, Shor [310] takes

$$\alpha_k = \alpha_0 q^k, \ 0 < q < 1. \tag{14.3.29}$$

But, such an α_k does not satisfy (14.3.15). For any given α_0 and q, as long as

$$\operatorname{dist}(x_1, S^*) > \frac{\alpha_0}{1-q}, \tag{14.3.30}$$

the sequence generated from the algorithm is not possible to close S^*.

The convergence result of the algorithm with step rule (14.3.29) is stated as follows.

Theorem 14.3.5 *Let $f(x)$ be convex and let there exist positive constant $\delta_1 > 0$ such that for all x,*

$$(x - x^*)^T g \geq \delta_1 \|g\| \|x - x^*\|, \ \forall g \in \partial f(x), \tag{14.3.31}$$

then there must exist constants $\bar{q} \in (0, 1)$ and $\bar{\alpha} > 0$ such that, provided that

$$q \in (\bar{q}, 1), \ \alpha_0 > \bar{\alpha}, \tag{14.3.32}$$

then the sequence $\{x_k\}$ generated by Algorithm 14.3.1 satisfies

$$\|x_k - x^*\| \leq M(\delta, \alpha_0) q^k, \tag{14.3.33}$$

where $x^* \in S^*$, $\bar{q}$ and $\bar{\alpha}$ are constants related to $\|x_1 - x^*\|$ and δ_1, $M(\delta_1, \alpha_0) > 0$ is a constant irrelative to k and related to δ_1 and α_0.

However, the rule (14.3.29) to determine stepsize is almost infeasible in practice, because, in general, it is impossible to know the values of $\bar{\alpha}$ and $\bar{q}$. If the given α_0 is too small, then (14.3.32) is not satisfied; if α_0 is too big, then the algorithm converges very slowly.

When f^* is known in advance, let us set

$$\alpha_k = \lambda \frac{f(x_k) - f^*}{\|g_k\|},\ 0 < \lambda < 2. \tag{14.3.34}$$

The convergence theorem of Algorithm 14.3.1 with stepsize rule (14.3.34) is due to Polyak [255] and stated as follows.

Theorem 14.3.6 *Let $f(x)$ be convex and the set S^* be nonempty. If there exist positive numbers $\bar{c}$ and $\hat{c}$ such that*

$$\|g\| \leq \bar{c},\ \forall g \in \partial f(x), \tag{14.3.35}$$

$$f(x) - f^* \geq \hat{c}\, dist(x, S^*) \tag{14.3.36}$$

hold for all x satisfying $dist(x, S^*) \leq dist(x_1, S^*)$, then the sequence generated by Algorithm 14.3.1 with stepsize (14.3.34) converges to some $x^* \in S^*$, and there exists a positive constant M such that

$$\|x_k - x^*\| \leq M q^k, \tag{14.3.37}$$

where $q = (1 - \lambda(2-\lambda)\hat{c}^2/\bar{c}^2)^{1/2} < 1$.

The above discussion has shown that the improvements only in the stepsize rule cannot, in general, significantly accelerate convergence. Indeed, slow convergence is due to the fact that the gradient is almost perpendicular to the direction towards the minimum. There is a simple way of changing the angles between the gradient and the direction towards the minimum. This can be done by performing a space dilation technique, which, in fact, is a generalization of the variable metric method.

Now we describe the space dilation method as follows:

Algorithm 14.3.7 *(The space dilation method)*

Step 1. Given initial point $x_1, \alpha > 0, H_1 = \alpha I; k := 1$.

Step 2. Evaluate $g_k \in \partial f(x_k)$; find the stepsize $\alpha_k > 0$; set

$$x_{k+1} = x_k - \alpha_k H_k g_k / (g_k^T H_k g_k)^{1/2}. \tag{14.3.38}$$

Step 3. Choose $r_k > 0$ and $\beta_k < 1$. Set

$$H_{k+1} = r_k \left(H_k - \beta_k \frac{H_k g_k g_k^T H_k}{g_k^T H_k g_k} \right). \tag{14.3.39}$$

$k := k+1$, *go to Step 2.* $\square$

It is not difficult to see that the matrix sequence $\{H_k\}$ generated by (14.3.39) are positive definite. There are various ways to choose α_k, β_k and r_k, for example,

$$\alpha_k = \frac{1}{n+1}, \ \beta_k = \frac{2}{n+2}, \ r_k = \frac{n^2}{n^2-1}. \tag{14.3.40}$$

Below, we state the convergence of the space dilation method without proof. The interested reader can consult Shor [311].

Theorem 14.3.8 *Let $f(x)$ be convex and the set S^* be nonempty. If*

$$dist(x_1, S^*) \ \le \alpha,$$

then the sequence $\{x_k\}$ generated by Algorithm 14.3.7 with (14.3.40) satisfies

$$\liminf_{k\to\infty} \frac{f(x_k) - f^*}{q^k} < +\infty, \tag{14.3.41}$$

where

$$q = \left(1 - \frac{2}{n+1}\right)^{\frac{1}{2n}} \frac{n}{\sqrt{n^2-1}}. \tag{14.3.42}$$

There are other generalizations to the subgradient method, for example, ellipsoid algorithm, finite difference approximation etc. We refer the readers to Zowe [386] and Shor [313] for details.

14.4 Cutting Plane Method

The cutting plane method for convex programming was presented independently by Kelley [186] and Cheney and Goldstein [58] respectively. The underlying idea of the cutting plane method is to find the minimum of a function on a convex polyhedral set in each iteration. After each iteration, a cutting plane is introduced, and a point, which does not satisfy the new hyperplane, is cut off from the feasible region, and hence the polyhedral set is reduced. At last, the iterates converge to a solution. The procedure is performed by solving a sequence of approximating linear programming.

For convex function $f(x)$, obviously, we have

$$f(x) = \sup_{y} \sup_{g \in \partial(y)} [f(y) + g^T(x-y)]. \tag{14.4.1}$$

Therefore, the minimization of $f(x)$ is equivalent to the following problem

$$\min \quad v \tag{14.4.2}$$
$$\text{s.t.} \quad v \geq f(y) + g^T(x-y),\ \forall y \in R^n, g \in \partial f(y). \tag{14.4.3}$$

The cutting plane method is just, at each iteration, to solve an approximation problem to (14.4.2)–(14.4.3). Let $x_i\ (i = 1, \cdots, k)$ be existing iterates. At each iteration, we would like to solve the subproblem

$$\min \quad v \tag{14.4.4}$$
$$\text{s.t.} \quad v \geq f(x_i) + g_i^T(x - x_i),\ i = 1, \cdots, k. \tag{14.4.5}$$

Obviously, the linear programming problem (14.4.4)–(14.4.5) is an approximation to problem (14.4.2)–(14.4.3).

We can state the cutting plane method as follows.

Algorithm 14.4.1 *(Cutting plane method)*

Step 1. Given an initial point $x_1 \in S$, where S is a given polyhedral set. Set $k := 1$.

Step 2. Compute $g_k \in \partial f(x_k)$.

Step 3. Solve the linear program (14.4.4)–(14.4.5) for v_{k+1} and x_{k+1}. Set $k := k+1$, go to Step 2. □

As indicated above, at each iteration, the algorithm adds a new constraint, which means, in geometry, that a part in S which does not contain the solution, will be cut off by a hyperplane.

The convergence of the cutting plane method can be stated below.

Theorem 14.4.2 *Let $f(x)$ be convex and bounded below. Then the sequences $\{x_k\}$ and $\{v_k\}$ generated by Algorithm 14.4.1 satisfy*

1) $v_2 \leq v_3 \leq \cdots \leq v_k \to f^*$.

2) Any accumulation point of $\{x_k\}$ is a minimizer of $f(x)$ in S.

Suppose that $f(x)$ is differentiable and the algorithm converges to a solution, then for k sufficiently large, $g_k = \nabla f(x_k)$ is very small, and hence the constraint condition (14.4.5) will be ill-conditioned. The other disadvantage of the cutting plane method is that when k is sufficiently large, there are too many constraints in problem (14.4.4)–(14.4.5) such that the cost is prohibitively expensive, since cutting plane constraints are always added to the existing set of constraints but are never deleted. Because of these disadvantages, the cutting plane methods have never been attractive, although it is one of the earliest methods for general convex programming. Therefore, some modified versions of the cutting plane methods are needed.

14.5 The Bundle Methods

The bundle method is a class of methods extended from the conjugate subgradient method. This is a descent method with $f(x_{k+1}) \leq f(x_k)$ for each k.

The conjugate subgradient method was presented by Wolfe [354]. At the k-th iteration, there is an index set $I_k \subset \{1, \cdots, k\}$. The search direction is determined by

$$d_k = -\sum_{i \in I_k} \lambda_i^{(k)} g_i, \; g_i \in \partial f(x_k), \tag{14.5.1}$$

where $\lambda_i^{(k)}$ $(i \in I_k)$ are obtained by solving the subproblem

$$\min \quad \left\| \sum_{i \in I_k} \lambda_i g_i \right\|_2^2 \tag{14.5.2}$$

$$\text{s.t.} \quad \sum_{i \in I_k} \lambda_i = 1, \; \lambda_i \geq 0. \tag{14.5.3}$$

When $f(x)$ is a convex quadratic function and $I_k = \{1, 2, \cdots, k\}$, under exact line search, the direction generated from (14.5.1)–(14.5.3) is the same as that of the conjugate gradient method. So, this method is said to be a conjugate subgradient method. We now state the algorithm as follows.

Algorithm 14.5.1 *(Conjugate Subgradient Method)*

Step 1. Given initial point $x_1 \in R^n$, compute $g_1 \in \partial f(x_1)$. Choose $0 < m_2 < m_1 < \frac{1}{2}$, $0 < m_3 < 1$; $\epsilon > 0$, $\eta > 0$, $k := 1$; $I_1 = \{1\}$.

Step 2. *Compute the direction* d_k *by (14.5.1)–(14.5.3).*
If $\|d_k\| \leq \eta$ *stop.*

Step 3. *Compute* $y_k = x_k + \alpha_k d_k$ *such that*

$$f(y_k) \leq f(x_k) - m_2 \alpha_k \|d_k\|_2^2, \tag{14.5.4}$$

or

$$\|y_k - x_k\| \leq m_3 \epsilon. \tag{14.5.5}$$

Step 4. *If there is* $g_{k+1} \in \partial f(y_k)$ *such that*

$$g_{k+1}^T d_k \geq -m_1 \|d_k\|_2^2, \tag{14.5.6}$$

then set $x_{k+1} := y_k$*, otherwise set* $x_{k+1} := x_k$*.*

Step 5. *Set* $I_{k+1} := I_k \cup \{k+1\} \setminus T_k$*, where* T_k *is an index set*

$$T_k = \{i \,|\, \|x_i - x_{k+1}\| > \epsilon\}.$$

Step 6. $k := k+1$*, go to Step 2.* □

The following convergence theorem was given by Wolfe [354].

Theorem 14.5.2 *Let* $f(x)$ *be convex and* $\|\partial f(x)\|$ *be bounded on some open set containing the set* $\{x \,|\, f(x) \leq f(x_1)\}$*. Let the sequence* $\{x_k\}$ *generated by Algorithm 14.5.1 make* $f(x_k)$ *bounded below. Then the algorithm must terminate in finitely many iterations.*

Now we consider an extension of the conjugate subgradient method. Suppose that we have performed several steps of the conjugate subgradient method. A certain number of points have been generated, at which the value of f has been computed together with some subgradient. We symbolize this information by the bundle $x_1, \cdots, x_k; f_1, \cdots, f_k; g_1, \cdots, g_k$; where $f_i = f(x_i)$ and $g_i \in \partial f(x_i)$.

Suppose that at k-th iteration we have weighted factors $t_i^{(k)} \geq 0$ $(i = 1, \cdots, k)$. Consider the following subproblem

$$\min \quad \left\| \sum_{i=1}^{k} \lambda_i g_i \right\| \tag{14.5.7}$$

$$\text{s.t.} \quad \sum_{i=1}^{k} \lambda_i = 1, \ \lambda_i \geq 0, \tag{14.5.8}$$

$$\sum_{i+1}^{k} \lambda_i t_i^{(k)} \leq \bar{\epsilon}, \tag{14.5.9}$$

where $\bar{\epsilon} > 0$ is a given constant. Write $\lambda_i^{(k)}$ as a solution of (14.5.7)–(14.5.9). Then the search direction of the bundle method is

$$d_k = -\sum_{i=1}^{k} \lambda_i^{(k)} g_i. \tag{14.5.10}$$

It is not difficult to see that if $t_i^{(k)} = 0\,(i \in I_k)$ and $t_i^{(k)} = +\infty\,(i \notin I_k)$, then (14.5.7)–(14.5.9) is equivalent completely to (14.5.2)–(14.5.3).

Algorithm 14.5.3 *(Bundle Method)*

Step 1. *Given initial point* $x_1 \in R^n$*, compute* $g_1 \in \partial f(x_1)$*. Choose* $0 < m_2 < m_1 < \frac{1}{2}$, $0 < m_3 < 1$, $\epsilon > 0, \eta > 0, k := 1$ *and* $t_1^{(1)} = 1$.

Step 2. *Solve (14.5.7)–(14.5.9) for* $\lambda_i^{(k)}$.
Compute d_k *by (14.5.10).*
If $\|d_k\| \leq \eta$ *stop.*

Step 3. *Compute* $y_k = x_k + \alpha_k d_k$ *such that (14.5.4) holds or*

$$f(y_k) - \alpha_k g_{k+1}^T d_k \geq f(x_k) - \epsilon, \tag{14.5.11}$$

where $g_{k+1} \in \partial f(y_k)$.
If (14.5.4) does not hold, then go to Step 5.

Step 4. $x_{k+1} := y_k, t_{k+1}^{(k+1)} = 1$,
$t_j^{(k+1)} = t_j^{(k)} + f(x_{k+1}) - f(x_k) - \alpha_k g_j^T d_k$, $j = 1, \cdots, k$.
Set $k := k + 1$*, go to Step 2.*

Step 5. $x_{k+1} := x_k, t_j^{(k+1)} = t_j^{(k)}\ (j = 1, \cdots, k)$
$t_{k+1}^{(k+1)} = f(x_k) - f(y_k) + \alpha_k g_{k+1}^T d_k$.
Set $k := k + 1$*, go to Step 2.* □

The convergence of the bundle method was established by Lemarechal [196] and stated below.

Theorem 14.5.4 *Under the assumptions of Theorem 14.5.2, Algorithm 14.5.3 will terminate in finitely many iterations, i.e., there exists $k \in IN$ such that $f(x_k) \le f^* + \epsilon$, where IN is an index set of positive integers.*

14.6 Basic Property of a Composite Nonsmooth Function

In the following two sections of the chapter, we will discuss a problem with the special form

$$\min_{x \in R^n} h(f(x)), \tag{14.6.1}$$

and develop the trust-region method for solving this class of problems. In (14.6.1), $f(x) = (f_1(x), \cdots, f_m(x))^T$ is a continuously differentiable function, and $h(f) : R^m \to R^1$ is convex but nonsmooth. The objective function in (14.6.1) is a composite function, and the problem (14.6.1) is referred to as composite nonsmooth optimization (for brief, composite NSO) or composite nondifferentiable optimization (for brief, composite NDO).

There are many examples of composite NSO in discrete approximation and data fitting. The following is a simple example.

Consider linear equations

$$Ax = b, \tag{14.6.2}$$

where $A \in R^{m\times n}$ and $b \in R^m$. If $m > n$, the equations (14.6.2) in general have no solution. However, we can take x such that the error between Ax and b is as small as possible. This means that we need to solve the minimization problem

$$\min_{x \in R^n} \|Ax - b\|, \tag{14.6.3}$$

where $\|\cdot\|$ is a norm on R^m. Obviously, (14.6.3) is a form of (14.6.1). If we take $\|\cdot\|_2$ in (14.6.3), the problem is just the classical least-squares problem.

In addition, note that a general smooth constrained optimization problem can be transformed to a composite NSO problem via an L_1 exact penalty function. This is the other reason that the composite NSO attracts us.

A prerequisite for describing algorithms for composite NSO is a study of optimality conditions for composite NSO, which is a direct use of the result in §14.1. For simplicity, we introduce the following conception:

$$\chi(x,d) = h(f(x)) - h(f(x) + A(x)^T d), \tag{14.6.4}$$

$$\psi_t(x) = \max_{\|d\|\le t} \chi(x,d), \tag{14.6.5}$$

$$DF(x,d) = \sup_{\lambda\in\partial h(f(x))} d^T A(x)\lambda, \tag{14.6.6}$$

where $\partial h(f(x))$ denotes the subgradient of $h(\cdot)$ at $f(x)$, $A(x) = \nabla f(x)^T$ is an $n\times m$ matrix.

Since $h(\cdot)$ is a convex function, by use of the chain rule of the subgradient of a composite function, it is not difficult to get the following lemma.

Lemma 14.6.1 *For composite function $\tilde f(x) = h(f(x))$, the fact*

$$0 \in \partial \tilde f(x) \tag{14.6.7}$$

is equivalent to

$$DF(x,d) \ge 0,\ \forall d \in R^n. \tag{14.6.8}$$

Then the stationary point of nonsmooth optimization satisfies (14.6.8). From the convexity of $h(f)$, we can also obtain the following results:

Lemma 14.6.2 *Let $\chi(x,d)$, $\psi_t(x)$, $DF(x,d)$ be defined in (14.6.4)–(14.6.6). Then*

1) $DF(x,d)$ exists for all x and d ;

2) $\chi(x,d)$ is a concave function with respect to d, its directional derivative at $d^*=0$ in the direction d is $-DF(x,d)$.

3) $\psi_t(x)\ge 0$, $\forall t\ge 0$; $\psi_1(x)=0$ if and only if x is a stationary point;

4) $\psi_t(x)$ is a concave function of t;

5) $\psi_t(x)$ is a continuous function of x for any given $t\ge 0$.

By the above results, we can show that the following statements are equivalent:

1) The sequence $\{x_k\}$ has an accumulation point x^* which is a stationary point.

2)

$$\liminf_{k\to\infty} \psi_1(x_k) = 0. \tag{14.6.9}$$

From the necessity theorem in §14.1, it follows that if x^* is a minimizer of $h(f(x))$, then it is a stationary point. For a special form of composite nonsmooth function, it can be written in the following equivalent form.

Theorem 14.6.3 *If x^* is a local minimizer of composite NSO problem (14.6.1), then there exists $\lambda^* \in \partial h(f(x^*))$ such that*

$$A(x^*)\lambda^* = 0, \tag{14.6.10}$$

where $A(x) = \nabla f(x)^T$.

Proof. It is enough to prove that (14.6.10) and

$$DF(x^*, d) \geq 0, \ \forall d \in R^n \tag{14.6.11}$$

are equivalent.

If (14.6.10) holds, then it follows from the definition (14.6.6) that (14.6.11) holds.

Now let us assume that (14.6.11) holds. Suppose to the contrary that (14.6.10) does not hold. Then the set

$$\bar{S} = \{A(x^*)\lambda \mid \lambda \in \partial h(f(x^*))\} \tag{14.6.12}$$

does not contain the origin. Since $\partial h(f(x^*))$ is a closed convex set, then $\bar{S}$ is too. Hence by applying the separation theorem of convex sets, we know there must exist $\bar{d} \in R^n$ such that

$$\bar{d}^T A(x^*)\lambda < 0, \ \forall \lambda \in \partial h(f(x^*)). \tag{14.6.13}$$

Since $\partial h(f(x^*))$ is closed, the above expression (14.6.13) contradicts the fact that $DF(x^*, \bar{d}) \geq 0$. This contradiction shows the equivalence between (14.6.11) and (14.6.10). $\Box$

Although the function $\tilde{f}(x) = h(f(x))$ may not be convex, we can obtain the following first-order sufficient conditions.

Theorem 14.6.4 *(First order sufficient conditions) If*

$$DF(x^*, d) > 0 \tag{14.6.14}$$

holds for all nonzero vectors d, then x^* is a strictly local minimizer of $h(f(x))$.

Proof. By (14.6.14), there exists $\delta > 0$ such that

$$DF(x^*, d) \geq \delta, \ \forall \|d\|_2 = 1. \tag{14.6.15}$$

Suppose that the theorem is not true, then there exists $x_k \to x^*$ with $h(f(x_k)) \leq h(f(x^*))$. Let us suppose that

$$x_k = x^* + \alpha_k d_k,\ \|d_k\|_2 = 1,\ \alpha_k > 0,\ \alpha_k \to 0_+.$$

Then

$$\begin{aligned}
& h(f(x_k)) - h(f(x^*)) \\
= \ & h(f(x^*) + A(x^*)^T(x_k - x^*)) - h(f(x^*)) + o(\alpha_k) \\
\geq \ & \alpha_k DF(x^*, d_k) + o(\alpha_k) \\
\geq \ & \alpha_k \delta + o(\alpha_k),
\end{aligned} \tag{14.6.16}$$

which contradicts the fact that $h(f(x_k)) \leq h(f(x^*))$. The contradiction proves the theorem. $\Box$

In fact, it also follows from (14.6.16) that, under assumption (14.6.14), there exist $\bar{\delta}$ and $\bar{\epsilon}$ such that

$$h(f(x)) - h(f(x^*)) \geq \bar{\delta}\|x - x^*\| \tag{14.6.17}$$

holds for all x with $\|x - x^*\| \leq \bar{\epsilon}$.

14.7 Trust Region Method for Composite Nonsmooth Optimization

For composite nonsmooth optimization (14.6.1), the subproblem of the trust-region method has the form

$$\min_{d\in R^n} \quad h(f(x_k) + A(x_k)^T d) + \frac{1}{2} d^T B_k d \triangleq \phi_k(d) \tag{14.7.1}$$

$$\text{s.t.} \qquad \|d\| \leq \Delta_k, \tag{14.7.2}$$

where $A(x) = \nabla f(x)^T \in R^{n\times m}$, $B_k \in R^{n\times n}$ is a symmetric matrix, and $\Delta_k > 0$ is a radius of the trust-region which is adjusted adaptively to be as large as possible subject to adequate agreement between $\phi_k(d)$ and $h(f(x_k + d))$ being maintained. The norm $\|\cdot\|$ in (14.7.2) is arbitrary but $\|\cdot\|_2$ is used in this section without special specification.

Let d_k be a solution of subproblem (14.7.1)–(14.7.2). Similar to Theorem 14.6.3, we can prove that there must exist

$$\lambda_k \in \partial h(f(x_k) + A(x_k)^T d_k), \tag{14.7.3}$$

$$\mu_k \in \partial \|d_k\|, \tag{14.7.4}$$

and $\bar{\mu}_k \geq 0$ such that

$$A(x_k)\lambda_k + B_k d_k + \bar{\mu}_k \mu_k = 0, \tag{14.7.5}$$

$$\bar{\mu}_k[\Delta_k - \|d_k\|] = 0. \tag{14.7.6}$$

The trust-region algorithm for composite nonsmooth optimization due to Fletcher [129] is as follows.

Algorithm 14.7.1 *(Trust-region algorithm for composite NSO)*

Step 1. Given $x_1 \in R^n, \lambda_0 \in R^m$, $\Delta_1 > 0$, $\epsilon \geq 0$, $k := 1$.

Step 2. Compute

$$B_k = \sum_{i=1}^{m} (\lambda_{k-1})_i \nabla^2 f_i(x_k); \tag{14.7.7}$$

Solve the subproblem (14.7.1)–(14.7.2) for d_k*;*

If $\|d_k\| \leq \epsilon$*, stop.*

Step 3. Calculate

$$r_k = \frac{h(f(x_k)) - h(f(x_k + d_k))}{\phi_k(0) - \phi_k(d_k)}. \tag{14.7.8}$$

If $r_k < 0.25$ *set* $\Delta_{k+1} := \|d_k\|/4$*;*
if $r_k > 0.75$ *and* $\|d_k\| = \Delta_k$*, set* $\Delta_{k+1} = 2\Delta_k$*;*
otherwise, set $\Delta_{k+1} = \Delta_k$*.*

Step 4. If $r_k > 0$ *go to Step 5;*
else $x_{k+1} := x_k$, $\lambda_k := \lambda_{k-1}$*, go to Step 6.*

Step 5. Set $x_{k+1} := x_k + d_k$, λ_k *is defined by (14.7.5).*

Step 6. $k := k + 1$*, go to Step 2.* □

To analyze the convergence of Algorithm 14.7.1, we assume that the sequence $\{x_k\}$ from the algorithm is bounded, which is implied if any level set $\{x \mid h(f(x)) \leq h(f(x_1))\}$ is bounded. The boundedness of $\{x_k\}$ suggests that there exists a bounded, closed convex set Ω such that

$$x_k \in \Omega,\ x_k + d_k \in \Omega,\ \forall k = 1, 2, \cdots. \tag{14.7.9}$$

Since $h(\cdot)$ is convex and well-defined on all of R^m, then there exists constant $L > 0$ such that

$$|h(f_1) - h(f_2)| \leq L\|f_1 - f_2\| \tag{14.7.10}$$

holds for all $f_1, f_2 \in f(\Omega) = \{v = f(x), x \in \Omega\}$. From the continuous differentiability of f and the boundedness of Ω, it follows that there is a constant $M > 0$ such that

$$\|A(x)\| \leq M \tag{14.7.11}$$

holds for all $x \in \Omega$.

Theorem 14.7.2 *Let $f_i(x)$ $(i = 1, \cdots, m)$ be twice continuously differentiable, if the sequence $\{x_k\}$ generated by Algorithm 14.7.1 is bounded, then there exists an accumulation point x^* of Algorithm 14.7.1, which is a stationary point of optimization problem (14.6.1).*

As to the proof of the theorem, please consult Fletcher (1981). Further, we have the following corollary.

Corollary 14.7.3 *Under the assumption of Theorem 14.7.2, if, instead of (14.7.7), $\|B_k\|$ is uniformly bounded, then $\{x_k\}$ has an accumulation point x^*, which is a stationary point.*

Now, the uniform boundedness of $\|B_k\|$ is relaxed to

$$\|B_k\| \leq c_5 + c_6 \sum_{i=1}^{k} \Delta_i. \tag{14.7.12}$$

Also, the adjustment of trust-region radius can be extended to the general case:

$$\|d_k\| \leq \Delta_{k+1} \leq \min[c_1\Delta_k, \bar{\Delta}], \quad \text{if } r_k \geq c_2, \tag{14.7.13}$$

$$c_3\|d_k\| \leq \Delta_{k+1} \leq c_4\Delta_k, \qquad \text{if } r_k < c_2, \tag{14.7.14}$$

where $c_i (i = 1, \cdots, 6)$ are positive constants and satisfy $c_1 > 1 > c_4 > c_3$, $c_2 < 1$; $\bar{\Delta}$ is a constant given in advance, an upper bound of the trust-region radius.

Under the extended conditions, we also can establish the convergence. We first give a lemma.

Lemma 14.7.4 *Let d_k be a solution of (14.7.1)–(14.7.2), then*

$$h(f(x_k)) - \phi_k(d_k) \geq \frac{1}{2}\psi_{\Delta_k}(x_k) \min\left[1, \frac{\psi_{\Delta_k}(x_k)}{\|B_k\|\Delta_k^2}\right], \tag{14.7.15}$$

where $\psi_t(x)$ is defined by (14.6.4)–(14.6.5).

Proof. It follows from the definition of d_k that

$$h(f(x_k)) - \phi_k(d_k) \geq h(f(x_k)) - \phi_k(d) \tag{14.7.16}$$

holds for any d with $\|d\| \leq \Delta_k$. By the definition (14.6.5) of $\psi_t(x)$, there exists $\|\bar{d}_k\| \leq \Delta_k$ such that

$$\psi_{\Delta_k}(x_k) = h(f(x_k)) - h(f(x_k) + A(x_k)^T \bar{d}_k). \tag{14.7.17}$$

Then, by using the convexity of $h(\cdot)$, we obtain that

$$\begin{aligned} h(f(x_k)) - \phi_k(d_k) &\geq h(f(x_k)) - \phi_k(\alpha\bar{d}_k) \\ &= \chi(x_k, \alpha\bar{d}_k) - \frac{1}{2}\alpha^2 \bar{d}_k^T B_k \bar{d}_k \\ &\geq \alpha\chi(x_k, \bar{d}_k) - \frac{1}{2}\alpha^2\|B_k\|\|\bar{d}_k\|^2 \\ &\geq \alpha\psi_{\Delta_k}(x_k) - \frac{1}{2}\alpha^2\|B_k\|\Delta_k^2 \end{aligned} \tag{14.7.18}$$

holds for all $\alpha \in [0, 1]$. Therefore

$$\begin{aligned} h(f(x_k)) - \phi_k(d_k) &\geq \max_{0\leq\alpha\leq 1}\left[\alpha\psi_{\Delta_k}(x_k) - \frac{1}{2}\alpha^2\|B_k\|\Delta_k^2\right] \\ &\geq \frac{1}{2}\min\left[\psi_{\Delta_k}(x_k), \frac{[\psi_{\Delta_k}(x_k)]^2}{\|B_k\|\Delta_k^2}\right]. \end{aligned} \tag{14.7.19}$$

We complete the proof. □

It is now possible to establish an extended conclusion of Theorem 14.7.2.

Theorem 14.7.5 *Let $f_i(x)\,(i = 1, \cdots, m)$ be twice continuously differentiable. Suppose that B_k in Algorithm 14.7.1 is not given by (14.7.7) but instead by (14.7.12) and that the sequence $\{x_k\}$ of the algorithm is bounded, then there must exist an accumulation point x^* of $\{x_k\}$ which is a stationary point of the problem (14.6.1).*

Proof. Suppose that the theorem does not hold, so there exists a positive constant $\delta > 0$ such that

$$\psi_1(x_k) \geq \delta, \ \forall k. \tag{14.7.20}$$

By use of 5) of Lemma 14.6.2, Lemma 14.7.4, inequality (14.7.20) and boundedness of Δ_k, we deduce that

$$\begin{aligned} h(f(x_k)) - \phi_k(d_k) &\geq c_7 \min\left[\Delta_k, \frac{1}{\|B_k\|}\right] \\ &\geq c_7 \min\left[\Delta_k, \frac{1}{c_5 + c_6 \sum_{i=1}^k \Delta_i}\right], \end{aligned} \tag{14.7.21}$$

where c_7 is a positive constant. Define a set

$$S = \{k \,|\, r_k \geq c_2\}, \tag{14.7.22}$$

then we have

$$\begin{aligned} h(f(x_1)) - \min_{x\in\Omega} h(f(x)) &\geq \sum_{k=1}^{\infty} [h(f(x_k)) - h(f(x_{k+1}))] \\ &\geq \sum_{k\in S} [h(f(x_k)) - h(f(x_{k+1}))] \\ &\geq c_2 \sum_{k\in S} [h(f(x_k)) - \phi_k(d_k)]. \end{aligned} \tag{14.7.23}$$

By (14.7.23), (14.7.21), (14.7.12) and $\Delta_k \leq \bar{\Delta}$, it follows that

$$\sum_{k\in S} \Delta_k \Big/ \left(c_5 + c_6 \sum_{i=1}^{k} \Delta_i\right) < +\infty. \tag{14.7.24}$$

In view of definition of Δ_{k+1}, we have

$$\Delta_{k+1} \leq c_4 \Delta_k, \ \forall k \notin S, \tag{14.7.25}$$

which gives

$$\sum_{i=1}^{k} \Delta_i \leq \left(1 + \frac{c_1}{1 - c_4}\right) \left[\sum_{\substack{i=1 \\ i\in S}}^{k} \Delta_i + \Delta_1\right]. \tag{14.7.26}$$

Combining (14.7.24) and (14.7.26) yields that $\sum_{i\in S} \Delta_i$ converges, and further that $\sum_{k=1}^{\infty} \Delta_k$ converges by (14.7.26) again. Hence $\|B_k\|$ is uniformly bounded. So, by Corollary 14.7.3, we know that (14.7.20) cannot hold for all k. The contradiction proves the theorem. □

Similar to the analysis of the trust-region method for unconstrained optimization, the condition (14.7.12) can further be weakened to

$$\|B_k\| \leq c_8 + c_9 k. \tag{14.7.27}$$

However, for the nonsmooth trust-region method, no matter what choices of B_k, there is only linear convergence. Several modifications are available to avoid the Maratos effect and enable the second-order rate to be established. The interested reader can consult Fletcher [131] and Yuan [369] for details.

14.8 Nonsmooth Newton's Method

Qi and Sun [283] extended the classical Newton's method to a non-smooth case by using the generalized Jacobian instead of the classical Jacobian. In this section, following Qi and Sun [283], we discuss the non-smooth Newton's method.

First, we introduce the generalized Jacobian and semismooth function. Suppose that $F : R^n \to R^m$ is a locally Lipschitzian function. Rademacher's theorem says that F is differentiable almost everywhere. Denote the set of points at which F is differentiable by D_F. We write $JF(x)$ for the usual $m \times n$ Jacobian matrix of partial derivatives whenever x is a point at which the necessary partial derivatives exist.

The generalized Jacobian of F at x, denoted by $\partial F(x)$, is a convex hull of all $m \times n$ matrices V obtained as the limit of a sequence of the form $JF(x_i)$, where $x_i \to x$ and $x_i \in D_F$. Then, we have

$$\partial F(x) = \text{co}\,\{\lim JF(x_i) \mid x_i \to x,\, x_i \in D_F\}. \tag{14.8.1}$$

Let F be Lipschitz on an open convex set U in R^n, and let x and y be points in U. Then, by Proposition 2.6.5 of Clarke [60], one has

$$F(y) - F(x) \in \partial F([x, y])(y - x). \tag{14.8.2}$$

Assume that for any $h \in R^n$,

$$\lim_{\substack{V \in \partial F(x+th) \\ t \downarrow 0}} \{Vh\} \tag{14.8.3}$$

exists. Then the classical directional derivative

$$F'(x;h) = \lim_{t\downarrow 0} \frac{F(x+th) - F(x)}{t} \tag{14.8.4}$$

exists, and

$$F'(x;h) = \lim_{\substack{V\in\partial F(x+th)\\ t\downarrow 0}} \{Vh\}. \tag{14.8.5}$$

In fact, by (14.8.2), we have

$$\frac{F(x+t_jh) - F(x)}{t_j} \in \operatorname{co}\partial F([x, x+t_jh])h.$$

By the Carathèodory theorem, there exist $t_j^{(k)} \in [0, t_j]$, $\lambda_j^{(k)} \in [0,1]$, $V_j^{(k)} \in \partial F([x, x+t_j^{(k)}h])$, for $k = 0, 1, \cdots, m$, $\sum_{k=0}^m \lambda_j^{(k)} = 1$, such that

$$\frac{F(x+t_jh) - F(x)}{t_j} = \sum_{k=0}^m \lambda_j^{(k)} V_j^{(k)} h.$$

By passing to a subsequence, we can assume that $\lambda_j^{(k)} \to \lambda_j$ as $j \to \infty$. We have $\lambda_j \in [0,1]$ for $k = 0, \cdots, m$ and $\sum_{k=0}^m \lambda_j = 1$. Then there are $t_j \downarrow 0$ such that

$$\begin{aligned} F'(x;h) &= \lim_{j\to\infty} \frac{F(x+t_jh) - F(x)}{t_j} = \lim_{j\to\infty} \{\sum_{k=0}^m \lambda_j^{(k)} V_j^{(k)} h\} \\ &= \sum_{k=0}^m \lim_{j\to\infty} \lambda_j^{(k)} \lim_{j\to\infty} \{V_j^{(k)} h\} = \sum_{k=0}^m \lambda_j \lim_{\substack{V\in\partial F(x+th)\\ t\downarrow 0}} \{Vh\} \\ &= \lim_{\substack{V\in\partial F(x+th)\\ t\downarrow 0}} \{Vh\}. \end{aligned}$$

F is called semismooth at x if F is locally Lipschitzian at x and

$$\lim_{\substack{V\in\partial F(x+th')\\ h'\to h, t\downarrow 0}} \{Vh'\} \tag{14.8.6}$$

exists for any $h \in R^n$. It implies that

$$\lim_{\substack{h'\to h\\ t\downarrow 0}} \frac{F(x+th') - F(x)}{t} = \lim_{\substack{V\in\partial F(x+th')\\ h'\to h, t\downarrow 0}} \{Vh'\}. \tag{14.8.7}$$

Lemma 14.8.1 *suppose that $F : R^n \to R^m$ is locally Lipschitzian and $F'(x;h)$ exists for any h at x. Then*
(1) $F'(x;h)$ is Lipschitzian;
(2) for any h, there exists a $V \in \partial F(x)$ such that

$$F'(x;h) = Vh. \tag{14.8.8}$$

Proof. For any $h, h' \in R^n$,

$$\|F'(x;h) - F'(x;h')\| = \left\| \lim_{t \downarrow 0} \frac{F(x+th) - F(x+th')}{t} \right\|$$
$$\leq \lim_{t \downarrow 0} \frac{\|F(x+th) - F(x+th')\|}{t} \leq L\|h - h'\|,$$

where L is the Lipschitzian constant near x. This proves (1).

By (14.8.2) and (14.8.4), there are a sequence $\{t_k\}$ and a sequence $\{V_k\}$ such that $t_k \downarrow 0, V_k \in \operatorname{co} \partial F([x, x + t_k h])$,

$$F'(x;h) = \lim_{k \to \infty} \{V_k h\}.$$

Because of the local Lipschitzian property of F, $\{V_k\}$ is bounded. By passing to a subsequence, we may assume that $V_k \to V$. Also since ∂F is closed, $V \in \partial F(x)$. So, (2) is proved. $\square$

If F is semismooth, then for any $V \in \partial F(x+h)$ and $h \to 0$,

$$Vh - F'(x;h) = o(\|h\|) \tag{14.8.9}$$

and

$$\lim_{\substack{x+h \in D_f \\ h \to 0}} \frac{F'(x+h;h) - F'(x;h)}{\|h\|} = 0. \tag{14.8.10}$$

In fact, if F is semismooth, we have a conclusion that the right-hand side of (14.8.7) is uniformly convergent for all h. Suppose that this conclusion does not hold. Then there exist $\epsilon > 0$, $\{h_k \in R^n \mid \|h_k\| = 1, k = 1, 2, \cdots\}$, $\|\bar{h}_k - h_k\| \to 0$, $t_k \downarrow 0$, $V_k \in \partial F(x + t_k \bar{h}_k)$ such that

$$\|V_k \bar{h}_k - F'(x;h_k)\| \geq 2\epsilon, \tag{14.8.11}$$

for $k = 1, 2, \cdots$. By passing to a subsequence, we may assume that $h_k \to h$. Thus, $\bar{h}_k \to h$ too. By Lemma 14.8.1 (1) and (14.8.11), we can get

$$\|V_k \bar{h}_k - F'(x;h)\| \geq \epsilon \tag{14.8.12}$$

for all sufficiently large k. This contradicts the semismoothness assumption.

The uniform convergence of the right-hand side of (14.8.7) implies the uniform convergence of the right-hand side of (14.8.5), which further implies (14.8.9).

Also, it immediately follows from (14.8.9) and (14.8.8) that (14.8.10) holds.

The Fréchet derivative $F'(x)$ is said to be strong if

$$\lim_{\substack{y\to x\\ z\to x}} \frac{F(z)-F(y)-F'(x)(z-y)}{\|z-y\|}=0. \tag{14.8.13}$$

Clearly, if F has strong Fréchet derivative at x, then F is semismooth at x.

If for any $V\in\partial F(x+h)$ and $h\to 0$,

$$Vh-F'(x;h)=O(\|h\|^{1+p}),$$

where $0<p\le 1$, then we call F p-order semismooth at x. Obviously, p-order semismoothness $(0<p\le 1)$ implies semismoothness.

Note that, if F is semismooth at x, then for any $h\to 0$,

$$F(x+h)-F(x)-F'(x;h)=o(\|h\|). \tag{14.8.14}$$

If F is p-order semismooth at x, then for any $h\to 0$,

$$F(x+h)-F(x)-F'(x;h)=O(\|h\|^{1+p}). \tag{14.8.15}$$

Now, we are in a position to give the nonsmooth Newton's method.

It is well-known that for smooth function $F: R^n\to R^n$, the Newton's method for solving the nonlinear equation

$$F(x)=0 \tag{14.8.16}$$

is

$$x_{k+1}=x_k-[F'(x_k)]^{-1}F(x_k). \tag{14.8.17}$$

Now, suppose that F is not a smooth function, but a locally Lipschitzian function. Then the formula (14.8.17) cannot be used. Let $\partial F(x_k)$ be the generalized Jacobian of F at x_k. Instead of (14.8.17), we may use

$$x_{k+1}=x_k-V_k^{-1}F(x_k), \tag{14.8.18}$$

where $V_k\in\partial F(x_k)$, to solve the nonsmooth equation

$$F(x)=0. \tag{14.8.19}$$

Lemma 14.8.2 *If all $V \in \partial F(x)$ are nonsingular, then there is a neighborhood $N(x)$ of x and a constant C such that for any $y \in N(x)$ and any $V \in \partial F(y)$, V is nonsingular and*

$$\|V^{-1}\| \leq C. \tag{14.8.20}$$

Proof. By contradiction. If the lemma is not true, there is a sequence $y_k \to x$, $V_k \in \partial F(y_k)$ such that either all V_k are singular or $\|V_k^{-1}\| \to \infty$. Since F is locally Lipschitzian, ∂F is bounded in a neighborhood of x. By passing to a subsequence, we may assume that $V_k \to V$. Then V must be singular, a contradiction to the assumption for this proposition. This completes the proof. □

Theorem 14.8.3 *(Local Convergence) Suppose that x^* is a solution of nonsmooth equation (14.8.19), F is locally Lipschitzian and semismooth at x^*, and all $V \in \partial F(x^*)$ are nonsingular. Then the iterative method (14.8.18) is well-defined and convergent to x^* in a neighborhood of x^*. If in addition F is p-order semismooth at x^*, then the convergence of (14.8.18) is of order $1 + p$.*

Proof. By Lemma 14.8.2, the iteration (14.8.18) is well-defined in the neighborhood of x^*. By (14.8.18), (14.8.9) and (14.8.14), we have

$$\begin{aligned}
\|x_{k+1} - x^*\| &= \|x_k - x^* - V_k^{-1}F(x_k)\| \\
&\leq \|V_k^{-1}[F(x_k) - F(x^*) - F'(x^*, x_k - x^*)]\| \\
&\quad + \|V_k^{-1}[V_k(x_k - x^*) - F'(x^*; x_k - x^*)]\| \\
&= o(\|x_k - x^*\|).
\end{aligned} \tag{14.8.21}$$

The case that F is p-order semismooth at x is similar. □

Finally we give the global convergence of nonsmooth Newton's method.

Theorem 14.8.4 *(Global Convergence) Suppose that F is locally Lipschitzian and semismooth on $S = \{x \in R^n \ : \ \|x - x_0\| \leq r\}$. Also suppose that for any $V \in \partial F(x)$ and $x, y \in S$, V is nonsingular,*

$$\|V^{-1}\| \leq \beta, \ \|V(y - x) - F'(x; y - x)\| \leq \gamma \|y - x\|,$$

and

$$\|F(y) - F(x) - F'(x; y - x)\| \leq \delta \|y - x\|,$$

where $\alpha = \beta(\gamma+\delta) < 1$ and $\beta\|F(x_0)\| \leq r(1-\alpha)$. Then the iterates (14.8.18) remain in S and converge to the unique solution x^* of (14.8.19). Moreover, the error estimate

$$\|x_k - x^*\| \leq [\alpha/(1-\alpha)]\|x_k - x_{k-1}\| \tag{14.8.22}$$

holds for $k = 1, 2, \cdots$

Proof. Obviously,

$$\|x_1 - x_0\| = \|V_0^{-1}F(x_0)\| \leq \beta\|F(x_0)\| \leq r(1-\alpha).$$

So $x_1 \in S$. Suppose now that $x_1, x_2, \cdots, x_k \in S$. Then

$$\begin{aligned}
\|x_{k+1} - x_k\| &= \|V_k^{-1}F(x_k)\| \leq \beta\|F(x_k)\| \\
&\leq \beta\|F(x_k) - F(x_{k-1}) - F'(x_{k-1}; x_k - x_{k-1})\| \\
&\quad +\beta\|V_{k-1}(x_k - x_{k-1}) - F'(x_{k-1}, x_k - x_{k-1})\| \\
&\leq \beta(\delta+\gamma)\|x_k - x_{k-1}\| = \alpha\|x_k - x_{k-1}\| \leq \alpha^k\|x_1 - x_0\| \\
&\leq r\alpha^k(1-\alpha).
\end{aligned} \tag{14.8.23}$$

Hence

$$\|x_{k+1} - x_0\| \leq \sum_{j=0}^{k} \|x_{j+1} - x_j\| \leq \sum_{j=0}^{k} r\alpha^j(1-\alpha) \leq r. \tag{14.8.24}$$

So $x_{k+1} \in S$, i.e., all the iterates (14.8.18) remain in S.

For any k and n,

$$\|x_{k+n+1} - x_k\| \leq \sum_{j=k}^{k+n} \|x_{j+1} - x_j\| \leq \sum_{j=k}^{k+n} r\alpha^j(1-\alpha) \leq r\alpha^k. \tag{14.8.25}$$

So the iterates (14.8.18) converge to a point x^* in S. Since F is Lipschitzian in S, $\|V_k\|$ is uniformly bounded. Thus

$$\|F(x^*)\| = \lim_{k\to\infty} \|F(x_k)\| \leq \lim_{k\to\infty} \|V_k\|\|x_{k+1} - x_k\| = 0,$$

i.e., $F(x^*) = 0$.

Suppose that there are $x^*, y^* \in S$ with $F(x^*) = 0$ and $F(y^*) = 0$. Let $V^* \in \partial F(x^*)$. Then

$$\begin{aligned}
\|y^* - x^*\| &\leq \beta\|V^*(y^* - x^*)\| \\
&\leq \beta\|V^*(y^* - x^*) - F'(x^*; y^* - x^*)\| \\
&\quad +\beta\|F(y^*) - F(x^*) - F'(x^*; y^* - x^*)\| \\
&\leq \beta(\delta + \gamma)\|y^* - x^*\| = \alpha\|y^* - x^*\|. \qquad (14.8.26)
\end{aligned}$$

This implies

$$\|y^* - x^*\| \leq 0,$$

i.e., $x^* = y^*$. This shows that x^* is the unique solution of (14.8.19).

Finally,

$$\begin{aligned}
\|x_{k+n+1} - x_k\| &\leq \sum_{j=k}^{k+n} \|x_{j+1} - x_j\| \leq \sum_{j=0}^{n} \alpha^{j+1}\|x_k - x_{k-1}\| \\
&\leq \frac{\alpha}{1-\alpha}\|x_k - x_{k-1}\|.
\end{aligned}$$

Setting $n \to \infty$, we obtain the result (14.8.22). $\square$

Exercises

1. Describe directional derivative, Dini directional derivative, Clarke directional derivative of f at x in the direction d respectively, and their properties and relations.

2. Describe the definition and properties of semi-smoothness.

3. Assume that $f(x)$ is continuously differentiable. Prove that

$$\partial f(x) = \nabla f(x).$$

4. Assume that $c_i(x)(i = 1, ..., m)$ are continuously differentiable. Let $f(x) = \max_{1\leq i\leq m} c_i(x)$ and $\bar{f}(x) = \sum_{i=1}^{m} |c_i(x)|$. Compute $\partial f(x)$ and $\partial \bar{f}(x)$.

5. Prove Theorem 14.3.3.

6. Assume that $f(x)$ is a convex function. Prove

$$f(x) = \sup_{y} \sup_{g \in \partial f(y)} [f(y) + g^T(x - y)].$$

7. Prove Theorem 14.4.2.

8. Prove the global convergence of the bundle method for uniformly convex functions.

9. Prove Lemma 14.6.2.

10. Apply the trust-region Algorithm 14.7.1 to problem

$$\min f(x) = \max\{1 + x_1 - x_2^2, 1 - x_1 + (1+\epsilon)x_2^2\}$$

where $\epsilon > 0$ is a small positive number with the starting point (δ, δ^2) and initial trust-region radius $\Delta_1 = 0.5\delta$, $\delta > 0$ being a small positive number. You should observe that the iterates converge only linearly if the trust-region is chosen $\{d| \quad \|d\|_\infty \leq \Delta_k\}$.

11. Prove Theorem 14.7.2.

12. Modify Algorithm 14.7.1 to derive a nonmonotone algorithm.

13. Give a generalized Newton's method for nonsmooth optimization and establish its global and local convergence.